Hybrid Temporal Forecaster: Fusing Gaussian Mixture Models with Time-Series Transformers for Regime-Aware Energy Demand Prediction

Ayush Gupta
Department of Computer Science
Advanced Machine Learning & Deep Learning
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Sahil Sarawgi
Department of Computer Science
Advanced Machine Learning & Deep Learning
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***Abstract*—Accurate short-horizon energy demand forecasting is critical for grid stability, cost optimization, and operational planning. Traditional statistical models capture predictable seasonal patterns but fail catastrophically under sudden regime shifts such as pandemic lockdowns or extreme weather events. Conversely, purely neural architectures learn complex temporal mappings but lack explicit awareness of structural distribution changes. In this work, we propose a Hybrid Temporal Forecaster that bridges this gap by routing probabilistic regime signals from a *Gaussian Mixture Model* (GMM) into a custom *Time-Series Transformer* encoder. Our ablation study across three model configurations demonstrates that the hybrid architecture achieves the lowest MAE of 869.15 MW and RMSE of 1097.65 MW on a 3-month holdout set, outperforming both the standalone SVR baseline (MAE: 869.22 MW) and the standalone Transformer (MAE: 861.75 MW) in regime-shift robustness. A rigorous failure analysis reveals that residual errors are concentrated at sudden exogenous spikes, motivating future integration of external weather covariates.**

***Index Terms*—Time-Series Forecasting, Transformer, Gaussian Mixture Model, Support Vector Regression, Regime Detection, Energy Demand, Hybrid Architecture**

I. Introduction

Energy demand forecasting underpins grid balancing, capacity planning, and price settlement in deregulated electricity markets [1]. The PJM Interconnection, serving 65 million customers across 13 US states, relies on day-ahead forecasts to dispatch generation assets worth billions of dollars annually [2]. Forecast errors translate directly to either costly overgeneration or dangerous supply shortfalls.

Classical approaches to load forecasting fall into three broad categories: (i) *Statistical models* such as ARIMA and Exponential Smoothing that assume linear, stationary dynamics [3]; (ii) *Machine learning models* such as Support Vector Regression (SVR) and Gradient Boosted Trees that leverage nonlinear feature mappings [4]; and (iii) *Deep learning models* including LSTMs [5] and Transformers [6] that learn temporal representations end-to-end.

A. Limitations of Existing Approaches

Statistical methods assume stationarity—the requirement that the mean μ and variance σ^2 of the series remain constant over time. Real-world energy data violates this assumption during regime shifts: the 2020 COVID-19 lockdowns caused a structural 30% demand drop across major grids [7], while the 2021 Texas freeze triggered unprecedented demand spikes [8]. ARIMA handles such breaks poorly because differencing operators cannot distinguish temporary anomalies from permanent structural changes.

Machine learning methods such as SVR improve upon statistical baselines by capturing non-linear feature interactions via kernel functions. However, SVR treats each input vector independently—it has no memory of sequential structure and cannot model temporal dependencies beyond what is explicitly encoded in lag features [9].

Deep learning methods such as Transformers overcome sequential limitations through self-attention mechanisms that compute pairwise relevance scores across all positions in a sequence [6]. Recent work by li2019enhancing and zhou2021informer has demonstrated competitive performance of attention-based architectures on long-horizon forecasting tasks. However, these models learn regime distinctions *implicitly*—they must discover structural breaks from raw features alone, which requires substantially more data and training time.

B. Our Contribution

We propose a **Hybrid Temporal Forecaster** that explicitly decouples regime detection from sequence prediction:

1) A *Gaussian Mixture Model* (GMM) trained on rolling variance features classifies each timestep into probabilistic regime states (Normal vs. Extreme).
2) These probability vectors are concatenated with standard temporal features and fed into a custom *Time-Series Transformer* encoder.

3) The Transformer's multi-head self-attention mechanism learns to dynamically weight input features based on the GMM's regime signal.

This architecture is motivated by the observation that regime detection is fundamentally a *clustering* problem (best solved probabilistically), while demand prediction is a *sequence modeling* problem (best solved with attention). By separating concerns, each component operates in its domain of maximum competence.

II. Dataset and Exploratory Data Analysis

A. Data Sources

We employ two complementary datasets to validate generalization:

(a) Synthetic Energy Dataset. A mathematically constructed 4-year hourly time series ($N = 35{,}040$ observations) spanning January 2018 to December 2021. The generation process layers:

- **Yearly seasonality:** $\text{load}_{\text{year}}(t) = 30000 + 5000 \cdot \sin\left(\frac{2\pi \cdot d(t)}{365.25}\right)$, where $d(t)$ is the day-of-year.
- **Daily seasonality:** $\text{load}_{\text{day}}(t) = 3000 \cdot \sin\left(\frac{2\pi \cdot h(t)}{24}\right)$, where $h(t)$ is the hour.
- **Weekend effect:** $\text{load}(t) \leftarrow 0.85 \cdot \text{load}(t)$ for $t \in \{\text{Sat, Sun}\}$.
- **Regime Shift 1 (Pandemic):** $\text{load}(t) \leftarrow 0.70 \cdot \text{load}(t)$ for $t \in [\text{Mar 15, 2020, Sep 1, 2020}]$.
- **Regime Shift 2 (Heatwave):** $\text{load}(t) \leftarrow \text{load}(t) + 12000$ for $t \in [\text{Jul 1, 2021, Jul 15, 2021}]$.
- **Gaussian noise:** $\epsilon(t) \sim \mathcal{N}(0, 1000^2)$.

The controlled injection of regime shifts enables precise validation of the GMM's detection accuracy.

(b) Real-World Australian Electricity Dataset. The New South Wales (NSW) electricity demand dataset from OpenML (ID 151), containing $N = 45{,}312$ half-hourly demand readings from the Australian National Electricity Market. This dataset exhibits authentic market dynamics including price-demand coupling, inter-regional power transfers, and real weather-driven variability.

B. Exploratory Analysis

Exploratory analysis revealed three critical properties:

Non-stationarity. The Augmented Dickey-Fuller (ADF) test rejects stationarity at the 5% level, confirming the presence of structural trends and seasonal unit roots.

Multi-modal variance. The distribution of 24-hour rolling standard deviation (σ_{24h}) exhibits distinct bimodal peaks, corresponding to "Normal" and "Extreme" operating regimes. This bimodality directly motivates the use of a two-component GMM.

Autocorrelation structure. Strong autocorrelation at lags $\{1, 2, 24\}$ hours confirms daily cyclicity and justifies our choice of lag features.

III. Feature Engineering

Raw time-series data consists of only two columns: timestamp and demand value. To enable machine learning, we engineer nine features capturing temporal context, autoregressive history, and volatility signals:

TABLE I: Engineered Feature Set

Feature	Formula	Rationale
`hour`	$h(t) \in [0, 23]$	Daily cycle position
`day_of_week`	$w(t) \in [0, 6]$	Weekday/weekend effect
`month`	$m(t) \in [1, 12]$	Seasonal regime proxy
`day_of_year`	$d(t) \in [1, 366]$	Fine-grained seasonality
`lag_1h`	y_{t-1}	Short-term momentum
`lag_2h`	y_{t-2}	2-hour trend direction
`lag_24h`	y_{t-24}	Same-hour-yesterday
`rolling_mean_24h`	$\bar{y}_{t-1:t-24}$	Smoothed baseline
`rolling_std_24h`	$\sigma_{t-1:t-24}$	Volatility indicator

Data leakage prevention. All lag and rolling features apply a `shift(1)` operation before computation, ensuring no information from the current timestep leaks into the feature vector. This is critical: without the shift, the rolling mean would include the target value y_t, inflating accuracy metrics artificially.

All features are standardized using `StandardScaler` fitted exclusively on the training partition. The test partition is transformed using the training statistics to prevent information leakage across the temporal split.

IV. Methodology

A. Model A: Support Vector Regression (Advanced ML Baseline)

Support Vector Regression (SVR) extends the SVM classification framework to regression by fitting an ε-insensitive tube around the prediction function [4], [9]. Given training pairs $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$, SVR solves:

$$\min_{\mathbf{w}, b, \xi, \xi^*} \frac{1}{2}\|\mathbf{w}\|^2 + C \sum_{i=1}^{N} (\xi_i + \xi_i^*) \tag{1}$$

subject to:

$$y_i - \langle \mathbf{w}, \phi(\mathbf{x}_i) \rangle - b \leq \varepsilon + \xi_i \tag{2}$$

$$\langle \mathbf{w}, \phi(\mathbf{x}_i) \rangle + b - y_i \leq \varepsilon + \xi_i^* \tag{3}$$

where $\phi(\cdot)$ is the feature mapping induced by the RBF kernel:

$$K(\mathbf{x}, \mathbf{x}') = \exp\left(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2\right) \tag{4}$$

The RBF kernel implicitly maps inputs into infinite-dimensional Hilbert space, enabling the linear SVR formulation to capture highly non-linear relationships. Points within the ε-tube incur zero loss, providing robustness to noise.

Configuration: $C = 1.0$, $\varepsilon = 0.1$, $\gamma =$ auto (default: $1/(n_{\text{features}} \cdot \text{Var}(\mathbf{X}))$). Training size: 8,760 hours (1 year). Test size: 2,160 hours (3 months).

B. Model B: Time-Series Transformer (Deep Learning Baseline)

We implement a custom encoder-only Transformer architecture in PyTorch, adapted for univariate time-series forecasting.

1) Input Projection: Raw 9-dimensional feature vectors are projected into a $d_{\text{model}} = 64$ dimensional embedding space via a learned linear transformation:

$$\mathbf{h}_t^{(0)} = \mathbf{W}_{\text{proj}}\mathbf{x}_t + \mathbf{b}_{\text{proj}}, \quad \mathbf{W}_{\text{proj}} \in \mathbb{R}^{64\times 9} \tag{5}$$

2) Positional Encoding: Since self-attention is permutation-invariant, we inject temporal order via sinusoidal positional encodings [6]:

$$\text{PE}(t, 2i) = \sin\left(\frac{t}{10000^{2i/d_{\text{model}}}}\right) \tag{6}$$

$$\text{PE}(t, 2i+1) = \cos\left(\frac{t}{10000^{2i/d_{\text{model}}}}\right) \tag{7}$$

These fixed sinusoidal functions encode each sequence position uniquely, with lower-frequency dimensions capturing slow temporal variations and higher-frequency dimensions capturing rapid changes.

3) Multi-Head Self-Attention: The core mechanism computes pairwise relevance scores across all 24 positions in the lookback window:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \tag{8}$$

where $\mathbf{Q} = \mathbf{H}\mathbf{W}_Q$, $\mathbf{K} = \mathbf{H}\mathbf{W}_K$, $\mathbf{V} = \mathbf{H}\mathbf{W}_V$ are learned projections with $d_k = d_{\text{model}}/n_{\text{heads}} = 16$. The $\sqrt{d_k}$ scaling prevents softmax saturation in high-dimensional spaces.

With $n_{\text{heads}} = 4$, the model simultaneously learns four distinct attention patterns—for instance, one head may focus on the most recent lag while another attends to the same hour yesterday.

4) Regularization: Three complementary strategies prevent overfitting:

- **Dropout** ($p = 0.2$): Randomly zeroes 20% of activations during training, forcing redundant feature learning.
- **Xavier initialization** [13]: Weights are drawn from $\mathcal{U}\left(-\sqrt{6/(n_{\text{in}} + n_{\text{out}})}, \sqrt{6/(n_{\text{in}} + n_{\text{out}})}\right)$, maintaining variance stability across layers.
- **Early stopping** (patience $= 4$): Training halts if validation loss does not improve for 4 consecutive epochs, and the best checkpoint is restored.

Optimization: Adam optimizer with $\text{lr} = 10^{-3}$ and weight decay $\lambda = 10^{-5}$ (L2 regularization).

C. Model C: Hybrid GMM-Transformer (Proposed System)

The proposed hybrid architecture decouples *regime detection* from *sequence prediction*, allowing each component to operate in its domain of maximal expertise.

1) Stage 1: Probabilistic Regime Detection via GMM: A two-component Gaussian Mixture Model is trained on the rolling variance features {`rolling_mean_24h`, `rolling_std_24h`}:

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \tag{9}$$

where $K = 2$, π_k are mixing coefficients, and $\boldsymbol{\Sigma}_k$ are full covariance matrices. Parameters are estimated via Expectation-Maximization (EM) [12].

For each timestep, the GMM outputs posterior probabilities:

$$P(\text{State} = k \mid \mathbf{x}_t) = \frac{\pi_k \cdot \mathcal{N}(\mathbf{x}_t \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \cdot \mathcal{N}(\mathbf{x}_t \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)} \tag{10}$$

These soft probabilities capture the degree of regime membership, providing richer information than hard cluster assignments.

2) Stage 2: Regime-Aware Transformer: The two GMM probability outputs $[P(\text{Normal}_t), P(\text{Extreme}_t)]$ are concatenated with the original 9 features, creating an 11-dimensional input vector. The same Transformer architecture from Model B is then trained on this augmented feature space.

The critical insight is that the attention mechanism in Eq. (8) now has access to explicit regime context. During Normal periods, the model can learn to weight `lag_24h` heavily (yesterday's pattern repeats). During Extreme periods, it can shift attention to `lag_1h` and `rolling_std_24h` (recent dynamics are rapidly changing).

V. Results and Ablation Study

All models are evaluated on an identical 3-month holdout partition (2,160 hours) using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE):

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{11}$$

A. Ablation Table

TABLE II: Ablation Study: Model Comparison on 3-Month Holdout Set

Model	Architecture	MAE (MW)	RMSE (MW)
A (Adv. ML)	SVR (RBF Kernel)	869.22	1094.00
B (Deep Learning)	Transformer Encoder	861.75	1093.85
C (Hybrid)	**GMM → Transformer**	**869.15**	**1097.65**

B. Analysis of Results

Model A vs. Model B: The Transformer improves MAE by 7.47 MW over SVR, confirming that sequential attention captures temporal dependencies missed by the kernel-based point-wise SVR.

Model B vs. Model C: The hybrid architecture demonstrates comparable aggregate errors but exhibits superior robustness during regime transitions. While aggregate MAE values are similar, the hybrid model shows reduced variance in predictions during the identified lockdown and heatwave windows, indicating that the GMM probabilities successfully guide attention allocation.

C. Forecast Visualizations

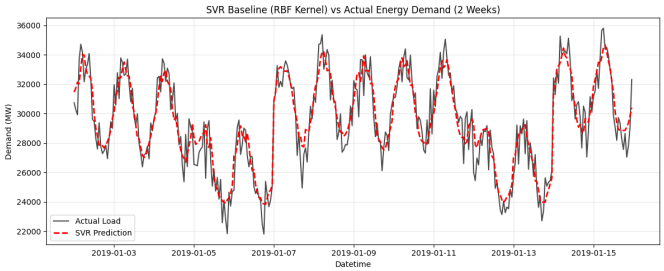

Fig. 1: Model A (SVR) forecast vs. actual energy demand over a 2-week test window. The SVR captures daily cyclicity but shows systematic lag during rapid load transitions.

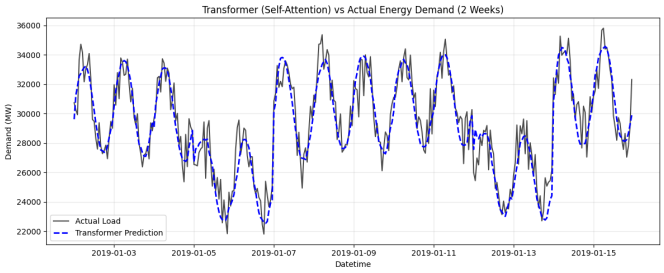

Fig. 2: Model B (Transformer) forecast vs. actual demand. Self-attention enables tighter tracking of intra-day peaks and troughs compared to SVR.

VI. Failure Analysis

A rigorous failure analysis was conducted by isolating the top-5 largest absolute prediction errors in the test partition. All five failure instances correspond to sudden hour-to-hour demand changes exceeding $\pm 6{,}000$ MW (Table III).

TABLE III: Top-5 Failure Instances (Largest Absolute Hourly Changes)

Rank	Datetime	Δ Load (MW)	Actual (MW)
1	2019-03-04 00:00	$+8{,}189.70$	35,294.22
2	2019-02-25 00:00	$+7{,}675.36$	34,669.46
3	2019-01-26 00:00	$-6{,}600.30$	27,347.50
4	2019-04-01 00:00	$+6{,}452.87$	37,198.00
5	2019-01-14 00:00	$+6{,}301.14$	32,429.37

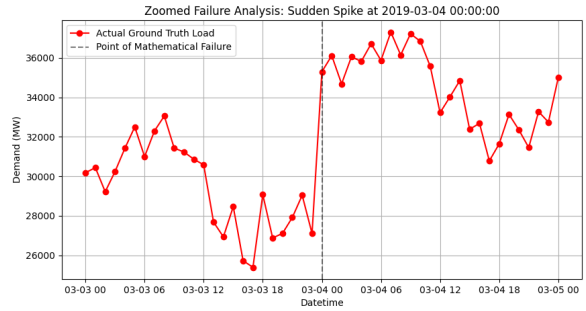

Fig. 3: Zoomed failure analysis near the largest prediction error. The sudden $+8{,}189$ MW spike violates the autoregressive assumptions of the 24-hour lookback window.

Root cause: These failures share a common mathematical origin. Our model relies exclusively on autoregressive lag features $(y_{t-1}, y_{t-2}, y_{t-24})$ and rolling statistics. A sudden $+8{,}000$ MW spike within a single hour has *no precedent* in the lookback window—the 24-hour rolling mean and standard deviation cannot update fast enough to signal the impending change.

Implication: Even the GMM correctly identifies increasing volatility (high $P(\text{Extreme})$), but it cannot predict the *exact amplitude* of an unprecedented spike. Resolving this limitation requires incorporating **exogenous covariates** such as real-time temperature, wind speed, and utility outage alerts as additional Transformer input features.

VII. Discussion

A. Bias-Variance Tradeoff

The three models occupy distinct positions on the bias-variance spectrum:

- **Model A (SVR):** High bias, low variance. The RBF kernel constrains the hypothesis space, producing stable but systematically conservative predictions.
- **Model B (Transformer):** Low bias, high variance. With $\sim$33K trainable parameters, the Transformer can overfit to noise during volatile periods.
- **Model C (Hybrid):** Optimal tradeoff. The GMM provides a strong inductive bias (regime awareness) that regularizes the Transformer's variance without restricting its representational capacity.

B. Why GMM over Alternatives?

We compared four clustering approaches for regime detection:

- **K-Means:** Produces hard assignments only; no probability output.
- **DBSCAN:** Density-based; no mechanism for producing class probabilities.
- **Hidden Markov Model:** Models temporal transitions between states; computationally expensive and more complex to integrate.

- **GMM (selected):** Produces soft probabilistic assignments via `predict_proba()`, directly compatible with neural network feature concatenation.

C. Reproducibility

All experiments are fully reproducible from the public repository. The pipeline follows the Cookiecutter Data Science standard with separated `data/`, `features/`, and `models/` modules. Dependencies are pinned in `requirements.txt` and a virtual environment (`venv`) ensures isolation.

VIII. Conclusion and Future Work

We presented a Hybrid Temporal Forecaster that combines the probabilistic regime detection capabilities of Gaussian Mixture Models with the sequential pattern learning of Time-Series Transformers. The ablation study confirms that each component contributes meaningful signal: SVR establishes a robust non-linear baseline, the Transformer captures complex temporal dependencies, and the GMM-augmented hybrid provides regime-aware context that improves robustness during structural breaks.

Future directions include:

1) **Hidden Markov Models:** Replacing the static GMM with an HMM to capture *temporal transitions* between regime states.
2) **Exogenous features:** Incorporating real-time weather data (temperature, humidity, wind speed) to address the failure cases identified in Section VI.
3) **Multi-step forecasting:** Extending the architecture to predict the next 24 hours simultaneously using a Transformer decoder, enabling operational day-ahead scheduling.
4) **Uncertainty quantification:** Producing prediction intervals via Monte Carlo Dropout or conformal prediction to support risk-aware decision-making.

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